

# CalVision

In particle physics, calorimetry refers to the detection of particles and measurement of their properties by the complete absorption of the particle's energy in a bulk of matter, referred to as a calorimeter.

Calvision is a consortium of universities and Department of Energy laboratories focused on advancing state-of-the-art calorimetric measurements for all types of particles with a higher level of precision. The program prioritizes the development of homogeneous calorimeters that maximize the use of available information. Key aspects of this program include harnessing dual readout of scintillation light signals and Cherenkov radiation (special radiation produced when a particle moves faster than the speed of light in that medium), utilizing timing to distinguish between these two types of light signals, and developing new algorithms to analyze these signals.

The current phase of the program is focused on developing an electromagnetic calorimeter that maximizes information usage, making it suitable for future lepton colliders such as the Future Circular Collider (FCC-ee). As part of this initiative, CUA's high-energy physics group plans to establish a crystal testing facility to characterize candidate crystals for use in the calorimeter. This classification involves studying the transmission spectrum, linear uniformity of light yield, decay time of scintillation signals, and the X-ray excited emission spectrum of the crystals. In this talk, I will discuss the methodology used for these studies, share some of the results I obtained for Bismuth Germanate (BGO) crystal samples, and explore the future scope of this work.